



Banking Customer & Risk Analytics

A Data-Driven View of Customers, Loans, Transactions and Branch Performance

Business Report

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Tools used: Python · MySQL · Power BI

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This report explains the project in plain language for both business and technical readers.

1. Executive Summary

This project turns raw banking records into a clear picture of how customers save, borrow and transact, and how bank branches perform across India. It combines a 5-table dataset covering 75,000 customers, 85,000 accounts, 30,000 loans, 500,000 transactions and 100 branches into one interactive Power BI dashboard, so that managers can answer everyday business questions in seconds instead of days.

In plain terms: the bank has ₹2.41 billion in deposits and ₹4 billion in active loans, spread across 75,000 customers. Most customers are in good financial health — only 5.39% of loans are in default, and two out of three loans are classed as low risk. Digital payments (UPI) now account for roughly half of all transactions, and a handful of branches bring in a disproportionate share of deposits.

The goal of this report is to explain, in plain language, what was built, what it found, and what the bank could do next.

2. Business Problem

Banks generate huge volumes of data every day — new accounts, loan repayments, card swipes, UPI transfers — but this data usually sits in separate systems and spreadsheets. Without a single, trustworthy view of it, leadership teams struggle to answer basic but important questions, such as:

- Which customers are most valuable, and which are at risk of leaving?
- Which types of loans are most likely to default, and why?
- Which branches and regions generate the most business?
- Are customers repaying loans comfortably, or are they financially stretched?

This project was built to solve exactly that problem: consolidate scattered banking data into one clean, reliable source, and present it through a dashboard that any manager — technical or not — can use to make faster, evidence-based decisions.

3. Project Scope

The project covers the full journey from raw data to business decision-making:

- Cleaning and validating five interlinked banking datasets.
- Answering 35+ real business questions using SQL (e.g. "which credit score band defaults the most?").
- Building calculated business metrics (KPIs) such as default rate, EMI-to-income ratio, and risk category.
- Designing a 4-page interactive Power BI dashboard for non-technical business users.

- Documenting insights and recommendations for decision-makers.

Out of scope: this project analyses historical, already-collected data. It does not connect to live banking systems, and it does not make automated lending decisions — it is a decision-support tool for human analysts and managers.

4. Data Sources

The analysis is built on five related datasets, structured the way a real bank's records typically are — one table per business object, linked by ID fields:

Dataset	Records	What it contains
Customers	75,000	Demographics: age, gender, income, occupation, state, customer segment
Accounts	85,000	Account type (savings, checking, credit card, loan account, fixed deposit) and balance
Loans	30,000	Loan type, amount, credit score, EMI, default status
Transactions	500,000	Payment mode, merchant, amount, date, linked account
Branches	100	Branch location, region, size, and deposits

Note: this is a structured, simulated dataset designed to resemble real retail-banking data, which allowed the full analytics workflow to be built and tested end-to-end without using sensitive real customer information.

5. Data Architecture

Data flows through three stages, each with a specific job — this keeps the analysis accurate, repeatable, and easy to update when new data arrives:

- Stage 1 — Raw CSV files: the original exports for customers, accounts, loans, transactions and branches.
- Stage 2 — Python: cleans and checks the data (fixes missing values, removes duplicates, standardises formats) and creates new business-friendly fields, e.g. age groups and income bands.
- Stage 3 — MySQL: the cleaned data is loaded into a relational database, where SQL queries answer specific business questions (e.g. "top 10 branches by deposits").
- Stage 4 — Power BI: connects to the results and turns them into an interactive dashboard that business users can filter and explore without writing any code.

The five tables are linked like this: a Customer can have several Accounts and Loans; each Account can have many Transactions; each Account and Loan is tied to a Branch. This relational structure means every number on the dashboard can be traced back to an individual customer, account or transaction.

6. Data Preparation

Before any analysis could be trusted, the raw data was cleaned and validated in Python. Key steps included:

- Checking for and handling missing or blank values (e.g. missing income or credit score).
- Removing duplicate customer, account and transaction records.
- Standardising formats — dates, currency values, and category labels (e.g. consistent loan type names).
- Creating new business fields for easier analysis, such as: age group, income group, EMI-to-income ratio, credit score category (Poor/Fair/Good/Very Good/Excellent), and high-risk vs low-risk loan flag.
- Validating relationships between tables so that every account, loan and transaction correctly links back to a real customer and branch.

This step is the foundation of the whole project — a dashboard is only as trustworthy as the data behind it, so this cleaning stage ensures every chart and KPI reflects accurate, consistent numbers.

7. Data Model

The five tables form a simple "one bank, many customers" structure, commonly known as a star-like relational model:

- Customers (1) → Accounts (many): one customer can hold several accounts (savings, credit card, fixed deposit, etc.)
- Customers (1) → Loans (many): one customer can have several active or closed loans
- Accounts (1) → Transactions (many): every transaction belongs to one account
- Branches (1) → Accounts / Loans (many): each account and loan is serviced by one branch

This structure allows the dashboard to answer questions at any level of detail — a single customer, a branch, a region, or the whole bank — simply by filtering, without needing to change any underlying calculations.

8. KPI Definitions

The dashboard is built around the following key performance indicators (KPIs). Plain-language definitions are given so non-technical readers can interpret every number correctly:

KPI	What it means
Total Customers	Total number of unique customers in the bank (75,000)

KPI	What it means
Total Deposits	Sum of all positive account balances (savings, checking, fixed deposit)
Total Loans / Loan Amount	Number and total value of loans issued
Default Rate	% of loans where the customer failed to repay as agreed
Transaction Value	Total ₹ value of all transactions processed
Average Transaction	Average ₹ value per transaction
Success Rate	% of transactions that completed successfully
EMI-to-Income Ratio	Monthly loan repayment (EMI) as a % of the customer's monthly income — a key affordability/risk signal
Credit Score Category	Groups customers into Poor, Fair, Good, Very Good, Excellent bands based on credit score
High-Risk vs Low-Risk Loan	A simplified 2-way risk flag combining credit score and repayment behaviour
Customer Segment	Platinum / Gold / Silver tiering used for relationship management

9. Dashboard Overview

The Power BI report has four pages, each targeted at a different business audience. All pages share the same filters (region, branch, state, gender, loan type) so users can drill down consistently.

9.1 Executive Overview

A single-page snapshot for leadership: total customers, transaction value, deposits, loans and default rate, plus monthly transaction trend, customer segment mix, top states by customer base, and loan status split.



Executive Overview — headline KPIs and trends for leadership

9.2 Customer Analytics

A demographic deep-dive: income group distribution, gender split, top occupations, and age-group breakdown — useful for marketing and relationship teams to understand who the bank's customers are.



Customer Analytics — who the bank's customers are

9.3 Loan & Risk Analytics

Focused on lending risk: average loan amount and EMI-to-income ratio by income group, loan type mix, and default rate by loan type — the primary page for credit risk teams.



Loan & Risk Analytics — where lending risk is concentrated

9.4 Branch & Transaction Analytics

Operational view for branch and payments teams: top merchants and payment modes, region-wise transaction value, and the top branches by deposits.



Branch & Transaction Analytics — operational performance by branch and region

10. Business Insights

Key findings from the analysis, explained in business terms:

10.1 Customers

- The customer base is fairly evenly split by gender (52% male, 48% female) and skews toward middle age — customers aged 30-59 make up roughly two-thirds of the base.
- Customer value is tiered into Platinum (43.8%), Gold (37.9%) and Silver (18.3%) segments — Platinum customers are the largest group, suggesting a generally affluent, loyal customer base.
- Maharashtra, Gujarat, Tamil Nadu and Karnataka together account for close to half of all customers, showing the bank's business is concentrated in a few key states.

10.2 Loans & Risk

- Home Loans dominate the loan book at 77.4% of all loans, followed by Business Loans (10.1%), Car Loans (5.6%) and Education Loans (3.1%) — the bank's lending risk is closely tied to the housing market.
- Overall default rate is a healthy 5.39%. Business Loans default slightly more often (5.64%) than Home Loans (5.23%) and Education Loans (5.03%).
- Customers with "Poor" credit scores default at 21.68% — roughly 4x the bank average — confirming credit score is a strong, reliable predictor of repayment risk.

- About two-thirds of loans (66.8%) are classified low risk versus one-third (33.2%) high risk, giving a quick portfolio health check.
- Lower-income customers carry a noticeably higher EMI-to-income burden (4.77%) than high-income customers (1.14%) — even though this is still a comfortable ratio overall, it is the group most sensitive to income shocks.
- Delhi (6.5%), Tamil Nadu (5.9%) and Rajasthan (5.8%) have the highest loan default rates by state, while West Bengal (4.5%) and Karnataka (4.9%) are the strongest performers.

10.3 Transactions & Payments

- UPI is by far the most-used payment method (49.8% of transactions), followed by Debit Card (20.2%) and Credit Card (14.9%) — confirming the shift toward instant digital payments.
- The North and East regions generate the highest transaction value (~₹1.45 billion each), while the West region trails at ~₹0.75 billion, highlighting a regional demand gap.
- Transaction success rate stands at 92.01%, meaning roughly 1 in 12 transactions currently fail or are declined — a potential area for payments-team follow-up.
- Deposits are not evenly spread — the top 5 branches (in Vijayawada, Ludhiana, Bhopal, Aurangabad and Agra) each hold ₹100M+, making them disproportionately important to the bank's funding base.

10.4 Accounts & Balances

- Fixed Deposits carry by far the highest average balance (~₹125K), followed by Savings (~₹38K) and Checking (~₹17K) accounts — showing customers prefer FDs for larger sums.
- Loan Accounts and Credit Cards naturally show negative average balances, since these represent money owed to the bank rather than deposited with it.
- New account openings were stable at roughly 7,300-7,470 per year from 2015 through 2025; the sharp drop shown for 2026 reflects only a partial year of data collected so far, not an actual decline.

11. Recommendations

Based on the findings above, the following actions are suggested for business teams:

- Tighten underwriting for "Poor" credit-score applicants, or price these loans with a risk premium, given their default rate is 4x the portfolio average.
- Introduce proactive outreach (repayment reminders, restructuring options) for lower-income customers whose EMI-to-income ratio is closest to the affordability limit.
- Investigate the 8% transaction failure rate — even small improvements in success rate at this volume represent meaningful value protected and a better customer experience.
- Study what the top-performing deposit branches (Vijayawada, Ludhiana, Bhopal, Aurangabad, Agra) are doing well, and test replicating those practices in underperforming branches.

- Prioritise growth investment in the West region, which lags materially behind other regions in transaction value.
- Continue to monitor Home Loan concentration (77% of the book) as a portfolio risk — diversifying into other loan types could reduce exposure to housing-market swings.
- Use the Platinum/Gold/Silver segmentation to design tiered retention and cross-sell offers, focusing on converting Silver customers upward.

12. Assumptions & Limitations

- The underlying dataset is a structured, simulated banking dataset built to resemble real retail-bank data; figures illustrate the analytics approach rather than a specific live institution.
- 2026 figures are partial-year (data available only through July), so year-over-year comparisons involving 2026 should be read with that in mind.
- "High risk" vs "low risk" loan classification is a simplified rule combining credit score and default history — it is a decision-support signal, not a regulatory or legally binding risk rating.
- The dashboard reflects a point-in-time snapshot of the data used to build it; it does not automatically refresh from a live banking system in its current form.
- Currency figures are presented in Indian Rupees (₹) unless otherwise noted.

13. Deployment Guide

How to run and use this project, written for both technical and business users.

13.1 For business users (view the dashboard)

- Open the .pbix file in Power BI Desktop (free to install from Microsoft), or view the published report via Power BI Service if it has been shared with you.
- Use the dropdown filters on the right of each page (region, branch, state, gender, loan type) to narrow the view to what you need.
- Click the navigation buttons at the top of the report to move between the four pages (Executive Overview, Customer Analytics, Loan & Risk Analytics, Branch & Transaction Analytics).

13.2 For technical users (reproduce the analysis)

- Clone the project repository from GitHub and install the Python requirements listed in requirements.txt.
- Run the Python cleaning/EDA scripts to regenerate the cleaned CSV files from the raw source data.
- Load the cleaned CSVs into MySQL using the provided schema, then run the SQL scripts to reproduce all 35+ business-question queries.

- Open the Power BI (.pbix) file and point its data source connections to your local MySQL instance, then click Refresh.

14. Future Enhancements

- Connect the pipeline to Azure SQL Database / Microsoft Fabric for scalable, cloud-hosted data storage.
 - Build a predictive loan-default model (machine learning) to flag high-risk applicants before a loan is approved, rather than only reporting on defaults after the fact.
 - Move from periodic refreshes to a near real-time dashboard fed by streaming transaction data.
 - Add customer churn prediction to proactively identify customers likely to close their accounts.
 - Introduce automated data-quality alerts so the team is notified if incoming data breaks expected patterns.
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